

Experiment 6

Aim: To demonstrate the performance of knn using wine dataset

Program

```
import pandas as pd
import pandas as pd
import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

wine=load_wine()

data=pd.DataFrame(data=wine.data,columns=wine.feature_names)

data['Target']=wine.target

x=data.drop('Target',axis=1)

y=data['Target']

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)

model=KNeighborsClassifier(n_neighbors=5)

model.fit(x_train,y_train)

y_pred=model.predict(x_test)

accuracy=accuracy_score(y_test,y_pred)

print("accuracy:",accuracy)

conf_matrix = confusion_matrix(y_test,y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='summer',
xticklabels=wine.target_names, yticklabels=wine.target_names)

plt.xlabel('Predicted Label')

plt.ylabel('True Label')
```

```
plt.title('Confusion Matrix')
```

```
plt.show()
```

Output:

